

Sheet G05 - Implicit Surfaces and Marching Cubes

Solutions for the theoretical and practical part via eCampus by Mo, 17.11.2025, 10:00.

Practical Part

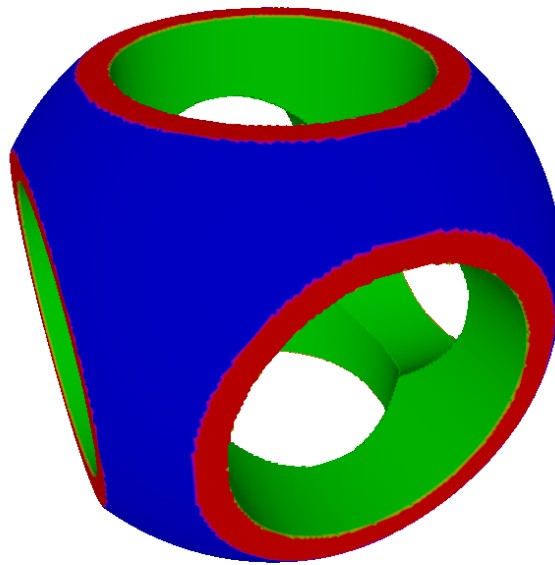


Figure 1: Marching Cubes.

Assignment 1) Marching Cubes

(5Pts)

Download the archive `framework_g05.zip` from eCampus and add it to the project like last time.

In the function `marching_cubes(...)`

- read the sdf and color values from the neighboring voxels.
- implement the computation of the marching cubes index.

In the function `interpolate_point(...)`

- implement the interpolation for the grid edges.

You can disable the grid in the settings menu or by setting `render_grid=false`; if you have performance problems.

Assignment 2) Constructive Solid Geometry

(4Pts)

In the same file as exercise 1, you can find definitions for `SDFSphere`, `SDFBox`, `SDFCylinder`, `SDFUnion`, `SDFIntersection` and `SDFDifference`.

- Implement the different shapes. You can replace `SDFHeart` in `onImGuiRender(...)` to test your implementation. For the cylinder, you do not have to consider the caps.
- Implement the SDF operators to combine different SDFs.
- Combine different SDFs (Sphere, Box, Cylinder) to achieve an image similar to Figure 1. Define the SDF in `onImGuiRender(...)`.

Hand in the solutions by uploading the `source.cpp` file.

Theoretical Part

Please hand in a sheet-G05-*lastname*.pdf sheet written in latex via eCampus!

Assignment 3) Gradient of Implicit Functions

(3Pts)

Let $\mathcal{M} = \{x \in \mathbb{R}^n \mid f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = 0\}$ be an implicit surface. Let $v \in \mathbb{R}^n$ be a vector. v is called a tangent vector of the surface $\mathcal{M}$ at the point x if and only if there exists a curve $\gamma : \mathbb{R} \rightarrow \mathcal{M}, \gamma \in \mathcal{C}^1$ such that $\gamma(0) = x \wedge \gamma'(0) = v$. Show that such a tangent vector is orthogonal to the gradient of f at this point.

Assignment 4) Marching cubes

(3Pts)

- Describe how the **marching cubes** algorithm works. Go through every step and describe how you can reconstruct a closed piecewise linear surface from an implicit representation
- What problems can you find using the **marching cubes** approach? Are there solutions to these problems?
- What are the main differences between **marching cubes** and **dual contouring**? Which one would you use in what situations?

Good luck!